

Chia-Yuan (Scott) Chang

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Applied Scientist, Amazon Rufus

Palo Alto, CA

I am looking for full-time roles in LLM post-training.

TL;DR

- **At Amazon Rufus (LLM post-training for reasoning)**

- I play a core role in LLM post-training for Amazon Rufus, focusing on SFT and RL for large-scale reasoning models.
- I built the full coding RLVR recipe on our 800B MoE model including data curation and filtering (29K verifiable coding problems), verifier and reward design, GSPO training on 256 H200, and evaluation — lifting LiveCodeBench-v6 pass@1 from 68.6% to 75.9% (+7.3%).
- I contributed to our general-agent RL recipe for in-house MoE models, including data collection and large-scale tool-use rollouts — improving MCP-Atlas pass@1 by +6.6 points.
- I developed our coding-agent RL pipeline for software-engineering tasks, including executable environment curation, sandboxed multi-turn rollouts with rewards, and large-scale RL training — improving SWE-bench Verified by +2.2 points.
- I was one of the main developers of our first reliable SFT training and data pipeline for our ~800B MoE model, driving major gains on reasoning and instruction-following benchmarks (IFEval +8%, AIME24 +25.5%, AIME25 +25%).
- I led in-house mid-training and was a key contributor to our long-context extension recipe for medium-sized LLMs.

- **Before and beyond current work**

- Built RAG systems with retrieval filtering and chunk selection, including *MAIN-RAG* (ACL 2025).
- Worked on efficient long-context LLM methods, including *Self-Extend* (ICML 2024 Spotlight).
- Developed research on domain generalization, fairness, healthcare ML, and time-series forecasting.

- **Selected recognition**

- CIKM 2023 Best Demo Paper Honorable Mention.
- 2nd and 4th Place Awards at TREC CAsT 2020.

Work Experience

Amazon, Rufus, Palo Alto, CA

Jun. 2025 – Present

- *Applied Scientist*
- Built the full coding RLVR recipe for our 800B MoE model — built a 29K verifiable-problem pool (execution-based rewards, model-based difficulty filtering, zero eval-set leakage) and trained with GSPO at large scale; LiveCodeBench-v6 pass@1 68.6% → 75.9% (+7.3%).
- Diagnosed and fixed reward-integrity failures at large scale (reward-grading starvation, truncation-driven response collapse); the response-budget fix turned a -4.4% decay into the +7.3% final gain.
- Lead and execute an in-house LLM CPT/mid-training.
- Core contributor to in-house LLM long-context extension recipe.

Amazon, Rufus, Palo Alto, CA

Nov. 2024 – Mar. 2025

- *Applied Scientist Intern*
- Researched state-of-the-art linearizing and hybrid LLM architectures.
- Proposed and implemented a new hybrid LLM architecture for internal exploration.

Visa Research, Foster City, CA

May 2024 – Aug. 2024

- *Research Intern*
- Developed a retrieval-augmented generation framework to identify and filter out unrelated chunks without model tuning.
- Explored business-facing application scenarios for RAG systems.

Texas A&M University, College Station, TX

Aug. 2021 – Dec. 2025

- *Graduate Research Assistant*

- Developed model-agnostic approaches for domain generalization and robustness.
- Built fairness-aware machine learning methods for healthcare applications.
- Studied efficient training and adaptation strategies for large language models.

Academia Sinica, Taipei, Taiwan

Oct. 2019 – Dec. 2020

- *Research Assistant*
- Developed ranking algorithms for financial news recommendation, reducing traders' daily reading time by 41%.
- Won 2nd place in an information retrieval competition with a T5-based coreference and reranking framework.

Selected Publications [\[Google Scholar\]](#)

Only first-author/selected publications are listed below; a complete publication list is available on Google Scholar.

Current record includes ACL, KDD, 2 ICML, NAACL, NeurIPS; 8 first-author papers; 800+ citations.

- [EACL2026]** **FaithLM: Towards Faithful Explanations for Large Language Models**
Y.N. Chuang, G. Wang, **C.Y. Chang**, et al., and X. Hu
The 19th Conf. of the European Chapter of the Association for Computational Linguistics (EACL), 2026
- [ACL2025]** **MAIN-RAG: Multi-Agent Filtering Retrieval-Augmented Generation**
C.Y. Chang, Z. Jiang, et al., and N. Zou
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025
- [KDD2025]** **CODA: Temporal Domain Generalization via Concept Drift Simulator**
C.Y. Chang, Y.N. Chuang, Z. Jiang, K.H. Lai, A. Jiang, and N. Zou
The 31st ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2025
- [ICML2024]** **LLM Maybe LongLM: Self-Extend LLM Context Window Without Tuning**
H. Jin, X. Han, J. Yang, Z. Jiang, Z. Liu, **C.Y. Chang**, H. Chen, and X. Hu
International Conference on Machine Learning (ICML), 2024
ICML'24 Spotlight (3.5%)
- [NAACL2024]** **Learning to Compress Prompt in Natural Language Formats**
Y.N. Chuang, S. Li, J. Yuan, G. Wang, K.H. Lai, L. Yu, S. Ding, **C.Y. Chang**, et al., and X. Hu
Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2024
- [NeurIPS2023]** **Auto-PINN: Understanding and Optimizing Physics-Informed Neural Architecture**
Y. Wang, X. Han, **C.Y. Chang**, D. Zha, U. Braga-Neto, and X. Hu
Thirty-seventh Conference on Neural Information Processing Systems (Workshop), 2023
- [AAAI2021]** **A Multi-step-ahead Markov Conditional Forward Model with Cube Perturbations for Extreme Weather Forecasting**
C.Y. Chang^{*}, C.W. Lu^{*}, and C.J. Wang
The 36th AAAI Conference on Artificial Intelligence, 2021
- [TREC2020]** **Query Expansion with Semantic-based Ellipsis Reduction for Conversational IR**
C.Y. Chang, et al.
The Twenty-Ninth Text REtrieval Conference (TREC), 2020
2nd place award in TREC'20

Open-Source Projects

- slimeⁿ**, Main Contributor 2026
- Main contributor to an open-source LLM RL training framework, enabling scalable multi-policy and student-teacher post-training workflows, <https://github.com/slime-n/slime-n>
- LongLM**, Contributor 2023
- Adapted self-extend implementation to open-source models, <https://github.com/datamllab/LongLM>

Education

- *Ph.D. in Computer Science*, Texas A&M University, College Station, TX
Advisors: Dr. Na Zou and Dr. Xia Hu
Aug. 2021 – Expected Aug. 2025
- *M.S. in Structures and Materials*, National Cheng Kung University, Tainan, Taiwan
Sep. 2013 – Jun. 2015